Homework: Text Analysis in Python

Topic: Files, Strings, Lists, Dictionaries, and Functions

Objective

In this homework, you will write a Python program that analyzes the words inside a text file.

This homework will help you practice:

- Opening and reading files
- Reading a file line by line
- Working with strings
- Using lists
- Using dictionaries
- Writing and calling functions

Problem Description

Write a Python program named:

`AnalyzeText.py`

Your program should ask the user to enter a file name, then open that file and analyze the words inside it.

The program should calculate:

- The total number of words in the file
- The number of unique words in the file
- The frequency of each word
- The 10 most frequent words
- The 10 longest words
- The 10 shortest words

Important Notes

- Convert all words to lowercase.
- Remove punctuation marks.
- Do not count numbers as words.
- Use functions to organize your program.
- You must open the file and read it line by line inside your code.

Required Functions

Your program must include the following functions:

```
def create_dictionary(filename):
    # Creates and returns:
    # total word count, unique word count, and word frequency
       dictionary
```

```
def most_frequent_words(word_dict, k):
    # Returns the top k most frequent words
```

```
def longest_words(word_dict, k):
    # Returns the top k longest words
```

```
def shortest_words(word_dict, k):
    # Returns the top k shortest words
```

```
def main():
    # Controls the program
```

Sample Output

```
Enter filename: sample.txt

Total words: 1200
Unique words: 450

Top 10 frequent words:
python - 25
program - 20
data - 18

Top 10 longest words:
communication
understanding
programming

Top 10 shortest words:
a
to
is
```

Starter Code

Complete the following Python code.

```python
# =========================================
# Homework: Text Analysis
# File: AnalyzeText.py
# =========================================

import string


# -----------------------------------------
# Function: create_dictionary
# -----------------------------------------
def create_dictionary(filename):

    # Dictionary for word frequencies
    word_dict = {}

    # Counters
    total_words = 0
    unique_words = 0

    # ---------------------------------
    # Open file
    # ---------------------------------
    file = open(filename, "r")

    # ---------------------------------
    # Read file line by line
    # ---------------------------------
    for line in file:

        # TODO:
        # Convert line to lowercase

        # TODO:
        # Remove punctuation

        # TODO:
        # Split line into words

        # TODO:
        # Process each word

            # TODO:
            # Ignore numbers

            # TODO:
            # Count total words

            # TODO:
            # Add/update word in dictionary
```

```
            # TODO:
            # Count unique words

    # --------------------------------
    # Close file
    # --------------------------------
    file.close()

    # TODO:
    # Return total words, unique words, dictionary

    pass


# ----------------------------------------
# Function: most_frequent_words
# ----------------------------------------
def most_frequent_words(word_dict, k):

    # TODO:
    # Sort dictionary by frequency

    # TODO:
    # Return top k words

    pass


# ----------------------------------------
# Function: longest_words
# ----------------------------------------
def longest_words(word_dict, k):

    # TODO:
    # Sort words by length, longest first

    # TODO:
    # Return top k longest words

    pass


# ----------------------------------------
# Function: shortest_words
# ----------------------------------------
def shortest_words(word_dict, k):

    # TODO:
    # Sort words by length, shortest first
```

```python
    # TODO:
    # Return top k shortest words

    pass


# ----------------------------------------
# Main Function
# ----------------------------------------
def main():

    # TODO:
    # Ask user for filename

    # TODO:
    # Call create_dictionary()

    # TODO:
    # Print total words

    # TODO:
    # Print unique words

    # TODO:
    # Print top 10 frequent words

    # TODO:
    # Print top 10 longest words

    # TODO:
    # Print top 10 shortest words

    pass


# ----------------------------------------
# Program Start
# ----------------------------------------
if __name__ == "__main__":
    main()
```

Submission Instructions

Submit only one file:

`AnalyzeText.py`

Make sure your program runs without errors before submission.